Examiner only

4. One type of car which is very efficient is a hybrid electric vehicle, which has both a conventional fuel engine and an electric motor which runs from batteries.

Data about a hybrid electric / petrol car is given below.

Minimum time taken to accelerate from 0 - 30 m/s	12 s
Mass of car	1 100 kg
Mean CO_2 emissions	90 g/km
Mean fuel economy	32 km/litre

(a) The car travels 160 km per week.

(i) Calculate the mean mass of CO_2 emitted by the car in a week. [1]

CO_2 = ... g

(ii) Calculate how many litres of fuel are used every week. [1]

Fuel used = ... litres

(b) (i) Use data from the table and an equation from page 2 to calculate the maximum acceleration of the car. [3]

Acceleration = ... m/s^2

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(ii) Use an equation from page 2 to calculate the size of the resultant force required to produce this acceleration. [2]

Resultant force = ... N

(c) State **one** other way cars are made more energy efficient. [1]

....

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